DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Examinatio[redacted] 2NOV 2017.

B. Tech Course in Electrical

Subject: - Electrical Utilisation **Semester: VIII**

Date: **Time: 3 Hours** **Max. Marks: 70**

Instructions to the students:

1. **Question no. 1 is compulsory and carry 10 marks whereas Question No. 2 to Question No. 7 carries 12 marks each**
2. **Illustrate you answers with neat sketches, diagrams etc. wherever required.**
3. **Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that part is a part of examination.**

If some part or parameter is noticed to be missing you may appropriately assume it and should mention it clearly.

Q.1] Solve the following: **(10)**

1] When load is above ...... a synchronous motor is found to be more economical?

a. 2 kW b. 20 kW c. 50 kW d. 100 kW

2] In jaw crushes a motor has to often start against .... load

a. Heavy b. medium c. Normal d. low

3] The process of depositing a coating of one metal over another metal or non- metal electrically called

a. Electro polishing b. Electro metallurgy c. Electro deposition d. None of these

4] Spot welding is used to weld metal pieces whose thickness

a. Greater than 12 mm b. Lesser than 12 mm c. Lies between 15 to 20 mm
d. Greater than 20 mm

5] .... is used for heating of non-conducting material

a. Eddy current heating b. Arc heating c. Induction heating d. Dielectric heating

6] Co-efficient of utilization depends upon....

a. colour of wall b. Colour of ceiling c. Size of room d. All of d above

7] Urban railways use

a. 750 V DC b. 440 V three phase AC
c. 220 V single phase AC d. 4.4 kV three phase AC

8] Pantograph collector is used in railways where the train runs at 100 to 130 kmph. Which among the following is true about pantograph collector?

a. It is unidirectional b. The erection of the overhead network is complicated
c. Its height cannot be varied d. None of these

9] The .... is the cheapest type of transport available in very dense traffic

a. tramway b. tram car c. trolley bus d. All of d above

10] Suburban railways use

a. 1500 V DC b. 440 V 3 phase AC c. 660 V 3 phase AC d. 3.3 kV 3 phase AC

Q. 2] Attempt the following. **(12)**

(a) What is electric heating? Name different methods of electric heating. Describe briefly the vertical core type induction furnace.

(b) State the main requirement for an ideal traction system. Name the different traction system. Give merits and de merits of electric traction over steam engine traction.

Q. 3] Attempt the following. **(12)**

(a) Suggest the reason the electric drive used for the following applications: i) rolling mills ii)lift, cranes iii) Coal mining iv) paper mill

(b) What is rheostatic braking? Explain the operation of **i)** DC **shunt** motor **ii)** DC series motor and when subjected to rheostatic braking. Show the torque speed characteristic during braking

Q. 4] Attempt any three of the following. **(12)**

(a) Define: **i)** solid angle **ii)** utilisation factor. **iii)** Maintenance factor **iv)** Depreciation factor

(b) What is electro plating? Explain in details?

(c) If 22.092 gram of nickel is deposited by 110 amp current following for 11 min, how much Cu would be deposited by 55 amp current in 7 min.? Automic weight of Ni and Cu are 58.6 and 63.18 respectively and valency of both is two.

(d) Compare AC welding and DC welding.

Q.5] Attempt the following. **(12)**

(a) A 3 phase, 50 kW, 4 pole, 980 rpm, induction motor has a constant load torque of 270Nm and at wide intervals additional torque of 1450Nm for 10 seconds. Calculate:
i) The moment of inertia of the flywheel used for load equalisation, if motor torque is not to exceed twice the rated torque.
ii)time taken after removal of additional load, before the motor torque becomes 630Nm

(b) A train runs between two stations 5 km apart at an average speed of 45km/hr. The run is to made according to simplified quadrilateral speed time curve. If the maximum speed to be limited to 80km/hr, acceleration 4km/hr/s, coasting retardation to 0.25 km/hr/s and braking retardation 3.5km/hr/s, determine duration of acceleration, coasting and braking periods.

Q. 6] Attempt the following. **(12)**

a) What is tractive effort of a train and what are its functions

b) Explain with the help of suitable circuit diagram.
i) Shunt transition ii) Bridge transition
As applied to a pair of dc traction motor

Q. 7] Write a short note on: **(12)**

(a) Trolley collector (b) Conductor rail system (c) Bow collector

*****All d Best*****